FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **One Section**: **Section-A.**

1. ***Section–A (01 – 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (08– 14, 26 – 32, 44 – 50) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section–A (15 – 18, 33 – 36, 51 – 54) *contains 6 paragraphs. Each paragraph is having 2 multiple choice questions having 4 options with* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

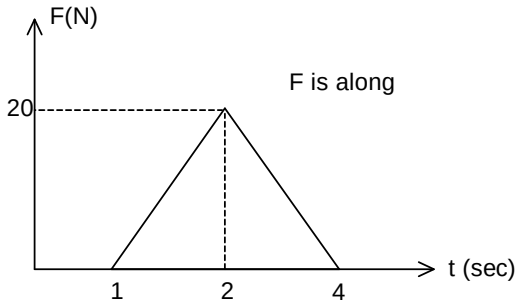
Physics

PART – I

SECTION – A
(One Options Correct Type)

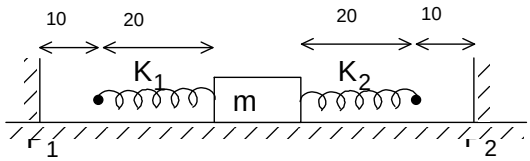
This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A particle of mass 2 kg is moving along x-axis with a velocity of 5 m/s. It crosses the origin at t = 0. A time varying force whose variation is as shown starts acting on a particle at t = 1 sec. For this situation, mark out the correct statement.



(A) the particle will come to rest instantaneously at t = 6.8 sec
(B) the final velocity of the particle will be 18 m/s.
(C) work done by the force F on the particle is 375 J.
(D) all of the above

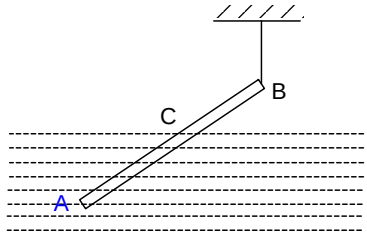
2. Two springs, each of unstretched length 20 cm but having different spring constants K_1 = 1000 N/m and K_2 = 3000 N/m are attached to two opposite faces of a block of mass m = 100 g kept on a smooth horizontal surface as shown in the figure. The outer ends of the two springs are now attached to two pins P_1 & P_2 whose locations are shown in the figure, as a result of this, the block acquires a new equilibrium position. The block has been displaced by small amount from its equilibrium position and released to perform simple harmonic motion, then



(A) new equilibrium position is at 35 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{100}$ sec.
(B) new equilibrium position is at 20 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{100}$ sec.
(C) new equilibrium position is at 35 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{26}$ sec.
(D) new equilibrium position is at 30 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{26}$ sec.

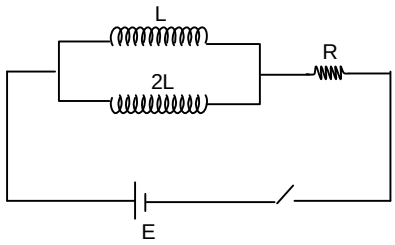
Space for Rough work

3. A uniform thin rod AB, 12 m long weighing 24 kg, is supported at end B by a flexible light string and a lead weight (of very small size) of 12 kg is attached at end A. The rod floats in a river with one-half of its length submerged. For this situation, mark out the correct statement. [Take g = 10 m/s^2, density of water = 1000 kg/m^3]



(A) the tension in the string is 36 g
(B) the tension in the string is 4 g
(C) the volume of the rod is 60000 m^3
(D) the point of application of the buoyancy force is passing through C (centre of mass of the rod).

4. Two ideal inductor of inductance L and 2L are connected in parallel. Combination of inductor are connected in series with a resistor of resistance R along with a battery of emf E. Switch S is closed at t = 0. After long time energy stored in inductor of inductance L will be?



(A) $\frac{LE^2}{2R^2}$ (B) $\frac{E^2}{R}$

(C) $\frac{LE^2}{9R^2}$ (D) $\frac{2LE^2}{9R^2}$

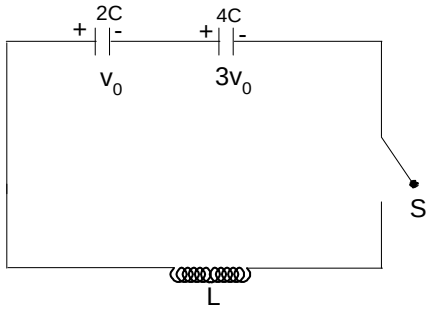
5. Radium $\left({}_{88}Ra^{226}\right)$ decays by emitting an α-particle into Rn^{222} with a half life of 1620 years, which then decays into Po^{218} by emitting another α-particle with a half life of 3.82 days. The ratio of Ra^{226} to Rn^{222} nuclei in a sample when Rn is in radioactive equilibrium:

(A) $1:1$ (B) $1.55\times10^5:1$

(C) $1.78\times10^5:1$ (D) $1.63\times10^5:1$

Space for Rough work

6. Two capacitors 2C and 4C initially charged to V_0 and $3V_0$ in the polarity shown are connected to an inductor of inductance 'L'. Initial current in the inductor is zero. Now the switch 'S' is closed. The maximum current in the circuit is



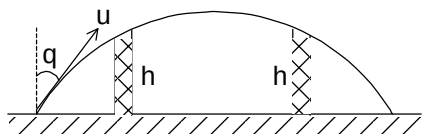
(A) $\frac{v_0}{8}\sqrt{\frac{C}{3L}}$

(B) $8V_0\sqrt{\frac{C}{3L}}$

(C) $\frac{v_0}{4}\sqrt{\frac{C}{3L}}$

(D) $4V_0\sqrt{\frac{C}{3L}}$

7. A particle is thrown with velocity u making an angle θ with the vertical. It just crosses the top of two poles each of height h after 1 sec & 3 sec respectively. The maximum height of projectile is



(A) 9.8 m (B) 19.6 m

(C) 39.2 m (D) 4.9 m

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

8. A t = 0, two particles B & C are located at the origin of the coordinate system. Then they start moving simultaneously. B moves under a constant acceleration of $2\hat{k}\,m/s^2$ with an initial velocity of $8\hat{j}\,m/s$. Particle C moves with constant velocity $\vec{V}_o$ in such a way that B & C collides at t = 4 sec. Then mark the incorrect statement(s).

(A) $\vec{v}_o = 8\hat{j} + 4\hat{k}\,m/s$

(B) position vector of location where two particles collide is $16\hat{i} + 32\hat{k}$ m.

(C) both (A) & (B) are correct

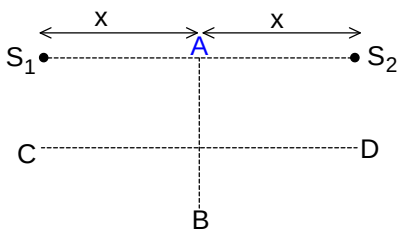
(D) It is not possible that B & C collide with each other for any value of $\vec{V}_o$

Space for Rough work

9. The gravitational potential changes uniformly from – 20 J/kg to – 40 J/kg as one moves along x-axis from x = -1 m to x = +1 m. Mark the correct statement about gravitational field intensity (E) at origin.
(A) E at x = 0 must be equal to 10 N/kg. (B) E at x = 0 may be equal to 10 N/kg.
(C) E at x = 0 may be greater than 10 N/kg. (D) E at x = 0 must not be less than 10 N/kg.

10. Mark the correct statement(s).
(A) when a sound wave strikes a wall, the compression pulse is reflected as compression pulse.
(B) when a sound wave strikes a wall, the compression pulse is reflected as rarefaction pulse.
(C) when a sound wave is coming out after passing through a narrow pipe, then reflection would be there at open end.
(D) when a sound wave is coming out after passing through a narrow pipe, then compression pulse is reflected as rare fraction pulse.

11. Two speakers are placed as shown in the figure below. Mark the correct statement(s).

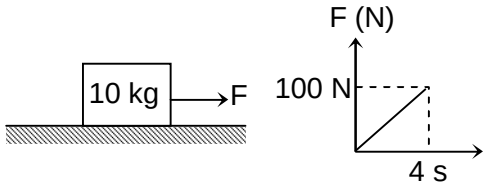


(A) if a person is moving along AB, he will hear sound loud, faint, loud and so on.
(B) if a person moves along CD, he will hear sound loud, faint, loud and so on.
(C) if a person moves along AB, he will hear maximum intensity sound which will continuously keep on decreasing
(D) if a person moves along CD, he will hear uniform intense sound.

12. A small sphere is charged uniformly and placed at some point A(x_0, y_0) so that at point B(9 m, 4m) electric field strength is $\vec{E}=(54\hat{i}+72\hat{j})NC^{-1}$ and potential is 1800 V. Then
(A) The magnitude of charge on the sphere is 4μC
(B) The magnitude of charge on the sphere is 2μC
(C) Coordinates of A are : $x_0 = -3$, $y_0 = -12$
(D) Coordinates of A are $x_0 = 4$, $y_0 = -1$

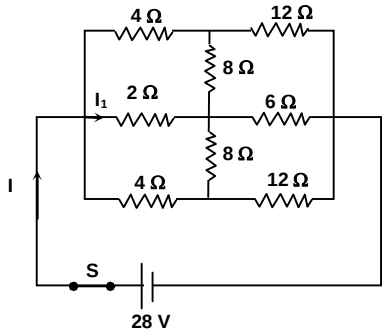
Space for Rough work

13. A 10 kg block is resting on a rough horizontal surface. A horizontal force F is applied to it for 4s. The variation of force with time is shown in the figure. ($\mu_s = \mu_k = 0.5$, g = 10 m/s^2). Then



(A) At t = 1 sec, velocity of block is 5 m/s
(B) At t = 1 sec, velocity of block is zero
(C) At t = 4 sec, velocity of block is 20 m/s
(D) At t = 4 sec, velocity of block is 5 m/s

14. In the circuit shown, the current
(A) I = 8 A
(B) I = 7 A
(C) I_1 = 3.5 A
(D) I_1 = 4 A

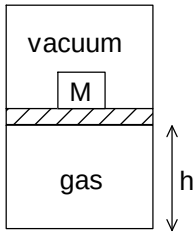


Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 15 and 16

One mole of an ideal diatomic gas is contained in the lower part of the cylinder beneath a piston of area 50 cm^2 as shown in the figure. The piston has negligible mass but a block of mass M = 100 kg is kept on the piston. Initially the gas is maintained at a temperature of 300 K and piston is at a height h from the bottom of the cylinder. The gas is now heated to raise the weight by 10 cm. [Take g = 10 m/s^2, atmospheric pressure = 10^5 N/m^2]



15. The thermal energy supplied to the gas in the process is
(A) 249 J (B) 348.6 J
(C) 99.6 J (D) 100 J

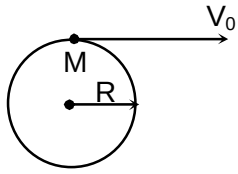
Space for Rough work

16. If a small hole has been created in the upper part of the cylinder, then the value of h before heating the gas is
(A) 1.66 m (B) 2.49 m
(C) 1.40 m (D) none of these

Paragraph for Question Nos. 17 and 18

Since gravitational field is conservative, we can conserve the total mechanical energy of the system of interacting particles. As the net force acting on a closed system is zero, we can conserve the linear momentum of the system. Since the gravitational forces between two interacting particles are action-reaction pair, the net torque experience by the system about any inertial point is zero. Hence, angular momentum of the system remains constant about any inertial point.

A body of mass m (<<M) is projected tangentially to earth's surface with a speed $v_0 = \sqrt{\frac{4GM}{3R}}$



17. The maximum height attained by the body is
(A) 2R (B) R
(C) 2/3 R (D) None of these

18. The areal velocity of the body at any instant is
(A) v_0R (B) $v_0R/2$
(C) $2v_0R/3$ (D) None of these

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Given: Enthalpy of formation of $H_2O(l)$ and $OH^-(aq.)$ are -285 kJ/mol and -227.3 kJ/mol respectively.
Enthalpy of neutralization of CH_3COOH with NaOH = -55 kJ/mol
The enthalpy of dissociation of CH_3COOH is:
(A) 2.3 kJ/mol (B) 2.7 kJ/mol
(C) 1.3 kJ/mol (D) 1.7 kJ/mol

20. Which salt is soluble in excess KCN as well as in excess KI solution?
(A) $Zn(NO_3)_2(s)$ (B) $Pb(NO_3)_2(s)$
(C) $AgNO_3(s)$ (D) $Hg_2(NO_3)_2(s)$

21. 20 ml of a mixture of C_2H_2 and CO was exploded with 30 ml of oxygen. The gases after the reaction had a volume of 34 ml. On treatment with KOH, 8 ml of oxygen gas remain unreacted. Which of the following options show a correct composition of the mixture?
(A) $\mathbf{V}_{C_2H_2}$ = 6 ml and V_{CO} = 14 ml (B) $\mathbf{V}_{C_2H_2}$ = 4 ml and V_{CO} = 16 ml
(C) $\mathbf{V}_{C_2H_2}$ = 2 ml and V_{CO} = 18 ml (D) $\mathbf{V}_{C_2H_2}$ = 5 ml and V_{CO} = 15 ml

22. Consider the cell: $Ag(s)/AgBr(s)/Br^-(aq)\,||\,Cl^-(aq)/AgCl(s)/Ag(s)$ at 298 K. The solubility product of AgBr and AgCl are 8×10^{-13} and 10^{-10} respectively. For what ratio of the concentrations of Br^- and Cl^- ions would the emf of the cell be zero?
(A) 7×10^{-3} (B) 8×10^{-3}
(C) 8×10^{-3} (D) None of these

Space for Rough work

23. Which of the following does not involve the role of H – bonding?
(A) Acetic acid dimerises in gaseous state
(B) In $NaHCO_3$, HCO_3^- ions are linked into an infinite chain
(C) 2nd ionisation constant in maleic acid is less than 2nd ionisation constant in fumaric acid
(D) Tertiary amine has the lowest boiling point than its isomeric primary and secondary amine

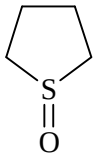
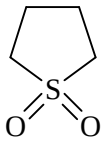
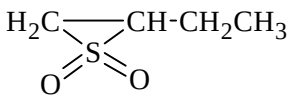
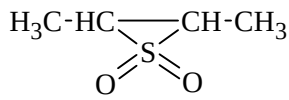
24. XY crystallizes in a bcc lattice in which the radius of $X^+ = 0.4\,\text{Å}$. If all the X^+ ions are removed and another cation Z^+ is inserted, what should be the maximum radius of the cation Z^+ to crystallize in a fcc lattice with Y^-?

(A) $0.548\,\text{Å}$ (B) $0.010\,\text{Å}$
(C) $0.400\,\text{Å}$ (D) $0.226\,\text{Å}$

25. 'Sulpholane' is an industrial solvent. The reaction sequence of its preparation is given below

1, 3-butadiene + SO_2 $\xrightarrow{100^oC}$ Addition product $\xrightarrow{H_2/Ni}$ Sulpholane

The structure of sulpholane is

(A) 
(B) 
(C) 
(D) 

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

26. The correct statement(s) regarding defects in solids is/are
(A) Frenkel defect is usually favoured by small difference in sizes of cations and anions
(B) Frenkel defect is a dislocation effect
(C) Paramagnetic substances are repelled by magnetic field
(D) Schottky defect has no effect on the physical properties of solids

Space for Rough work

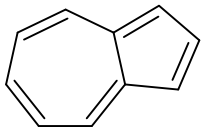
27. Which of the following is/are ortho & para directing for incoming electroiphile?
(A) Ph-SOEt (B) Ph-N=O
(C) Ph-CH=CH-CHO (D) Ph-CH=CH_2

28. Which of the following reactions are giving correct results?
(A) $NH_2-CO-NH_2 \xrightarrow{Br_2+KOH} NH_2-NH_2$
(B) $Ph-CO-COOH \xrightarrow{\text{conc. } H_2SO_4/\Delta} Ph-COOH+CO$
(C) $(R)-CH_3-CH(Br)COONa \xrightarrow{NaOH} (R)-CH_3-CH(OH)COONa$
(D) $\text{Benzene}+HCN \xrightarrow{HCl/H_2O} \text{Benzamide}$

29. Which of the following statement(s) is/are CORRECT?
(A) At 369 K, both α and β forms of sulphur are stable to exist and this temperature is called transition temperature
(B) BiF_3 is predominantly covalent
(C) Diborane catches fire spontaneously upon exposure to air
(D) I_2O_5 is used in the estimation of carbon monoxide

30. Radionucleide 'P' decays into 'Q' and 'Q' decays into 'R' as:
$P \xrightarrow{t_{1/2}=1000 \text{ hrs}} Q \xrightarrow{t_{1/2}=10 \text{ min}} R$
Number of nuclei of Q and R after 1000 hrs, if we start with 2 moles of P, are [$N_A = 6 \times 10^{23}$]
(A) $N_Q = 6 \times 10^{23}$ (B) $N_Q = 10^{20}$
(C) $N_R = 6 \times 10^{23}$ (D) $N_R = 10^{23}$

31. Which of the following statement/s is/are correct regarding Azulene?



(A) Azulene is an isomer of naphthalene and is blue in colour
(B) Azulene has very high dipole moment while naphthalene has dipole moment zero
(C) Seven membered ring of azulene is electrophilic and five membered ring is nucleophilic
(D) Seven membered ring of azulene is nucleophilic and five membered ring is electrophilic

Space for Rough work

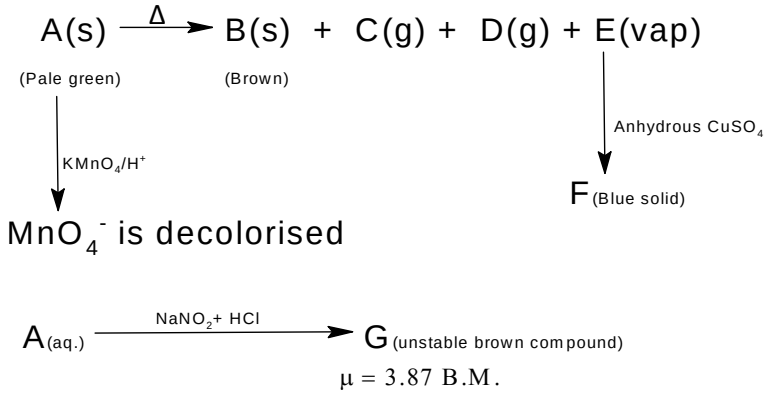
32. When $FeSO_4$ absorbs NO in aq. Solution, a new compound is formed. Select the correct statement about new compound.
(A) It is brown in colour
(B) It has d^2sp^3 hybridisation
(C) It has four unpaired electron
(D) Oxidation number of Fe is +1

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 33 and 34

Consider the following reaction(s)



33. Sum of oxidation state of the central atom in C(g) and D(g) is
(A) 6
(B) 8
(C) 10
(D) 4

34. Select INCORRECT statement about compound 'F'.
(A) It absorb orange-red light for d-d transition
(B) 4 water molecules are directly bonded with each Cu^{+2}
(C) It is paramagnetic
(D) It's dipole moment = $\sqrt{8}$ B.M.

Space for Rough work

Paragraph for Question Nos. 35 and 36

For the given reaction sequence:

$$\underset{(S)}{CH_3-CH_2-COOH} \xrightarrow{N_3H,H^+/\Delta} (P) \xleftarrow{Br_2/KOH} (Q) \xrightarrow{LiAlH_4} (R)$$

35. Choose the incorrect option.
(A) (P) and (R) are homologous to each other
(B) Basic strength order is: (Q) > (P) > (R)
(C) (P) and (Q) can be distinguished by carbyl amine test
(D) (Q) can be obtained by reacting (S) with NH_3, Δ

36. Choose the correct option.
(A) (P) is propyl amine
(B) (R) is ethyl amine
(C) (Q) is acetamide
(D) Schmidt reaction is involved in reaction sequence

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle, then 2r equals

(A) $\sqrt{PQ.RS}$ (B) $\frac{PQ+RS}{2}$

(C) $\frac{2PQ.RS}{PQ+RS}$ (D) $\sqrt{\frac{PQ^2+RS^2}{2}}$

38. If $f(x)=\frac{\sin^{-1}\left(\frac{x}{^{10}C_x}\right)+\cos^{-1}\left(\frac{x}{^{10}C_x}\right)}{\sqrt{5[x]^2+6\{x\}^2-60[x]+77\{x\}+160}}$, (where [.] denotes the GIF and {.} denotes the fractional part) then domain of f(x) is :

(A) $\{x \mid x \in I, 0 \le x \le 10\}$ (B) {0, 1, 2, 3, 4, 9}

(C) {0, 1, 2, 3, 4, 9, 10} (D) {0, 1, 2, 3, 9}

39. On 8×8 Chess board. If two squares of 1×1 are selected randomly, what is probability that, they have one side common.

(A) $\frac{1}{16}$ (B) $\frac{1}{18}$

(C) $\frac{1}{20}$ (D) $\frac{1}{22}$

40. A triangle ABC with vertices $A\equiv(2,5), B\equiv(5,2)$ and $C\equiv(-1,-1)$ inscribed on the curve $xy-x-y-3=0$ and $H\equiv(\alpha,\alpha)$ is orthocenter. Let D, E and F be the feet of perpendiculars from A, B and C on BC, CA and AB respectively and inradius of ΔDEF is r then

(A) Orthocenter of triangle ABC is (-1, -1) (B) Equation of DE is $x+y-\frac{13}{5}=0$

(C) $r=\frac{4}{5}\sqrt{2}$ units (D) $r=\frac{2}{5}\sqrt{2}$ units

Space for Rough work

41. Let PQ and QR be diagonals of adjacent faces of a rectangular box, with its centre at O. if $\angle QOR, \angle ROP$ and $\angle POQ$ are θ, ϕ and ψ respectively then the value of '$\cos\theta + \cos\phi + \cos\psi$' is

(A) –2 (B) $-\sqrt{3}$
(C) –1 (D) 0

42. For x > 0, let f(x) be the minimum value of $\left|X - \sqrt{m^2 + 2n^2}\right|$ for all integers m, n. Then $\lim\limits_{x\to\infty}$ f(x) is equal to

(A) 0 (B) 1
(C) 2 (D) 4

43. If $\int_0^{\frac{\pi}{2}} \frac{dx}{a + b\cos x} = \frac{1}{\sqrt{b^2 - a^2}} \ln \frac{b + \sqrt{b^2 - a^2}}{a}$, (where b > a), then the value of $\int_0^{\frac{\pi}{2}} \frac{\cos x}{(3 + 5\cos x)^2} dx$ is

(A) $\frac{1}{16} + \frac{5}{64}\ln 3$ (B) $\frac{1}{16} - \frac{5}{64}\ln 3$
(C) $-\frac{1}{16} + \frac{5}{64}\ln 3$ (D) $-\frac{1}{16} + \frac{64}{5}\ln 3$

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

44. If $k_1 = \tan 27\theta - \tan\theta$ and $k_2 = \frac{\sin\theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta}$, then

(A) $k_1 = 2k_2$ (B) $k_1 = k_2$
(C) $\frac{k_1 + k_2}{2k_1 - k_2} = 1$ (D) $\frac{k_1 + k_2}{2k_1 - k_2} = 2$

45. Consider a differentiable f : R $\to$ R for which $f'(0) = \ln 2$ and $f(x + y) = 2^x f(y) + 4^y f(x) \forall x, y \in R$. Which of the following is(are) correct

(A) f(4) = 240 (B) $f'(2) = 24 \ln 2$
(C) the minimum value of y = f(x) is $-\frac{1}{4}$ (D) the number of solution of $f(x) = 2$ is 1

Space for Rough work

46. ABCD is a rhombus. If M, N, P are points taken on sides AB, BC, CD respectively such that centroid of ΔMNP is G. If AM = $\frac{1}{12}$ units, BN = $\frac{1}{5}$ units then

(A) if G lies on AC, PD = $\frac{7}{60}$ units (B) if G lies on BD, PD = $\frac{31}{60}$ units

(C) if G lies of AC, PC = $\frac{13}{20}$ units (D) If G lies on BD, PC = $\frac{17}{60}$ units

47. A parabola C whose focus is S(0, 0) and passing through P(3,4) Equation of tangent at P to parabola is 3x + 4y -25 = 0. A chord through S parallel to tangent at P intersects the parabola at A and B

(A) length of AB is 20 units

(B) latus rectum of parabola is 20 units

(C) the point from which three real and coincident normal are drawn to parabola is (-3, ,-4)

(D) the point from which three real and coincident normal are drawn to parabola is (-6, -8)

48. Let M be a 3×3 orthogonal matrix. Let A be the 2×2 matrix in the upper – left corner of M and let B the 1×1 matrix in the lower – right corner of M. Let M_1 be the 3×3 matrix whose first 2 columns are first 2 columns of M and whose remaining 1 columns is the last column of I_3. Let M_2 be the 3×3 matrix whose first 2 columns identical to I_3 and the last 1 columns those of M. Which of the following is(are) correct?

(A) $MM_2 = M_1$ (B) $M^TM_2 = M_1$

(C) $\left|\det(M_1)\right| = \left|\det(M_2)\right|$ (D) $\left|\det(A)\right| = \left|\det(B)\right|$

49. Let $I = \int_0^{1/2}\left(\frac{1}{4}-x^2\right)^4 dx$. then

(A) $I = \int_0^{1/2} x^4(1-x)^4\,dx$ (B) $I = \frac{1}{2}\int_0^1 x^4(1-x)^4\,dx$

(C) $I = \int_0^{\pi/2} \sin^9\theta\cos^9\theta\,d\theta$ (D) $I = \frac{1}{1260}$

50. If $\left(\frac{x}{a}\right)+\left(\frac{y}{b}\right)=1$ and $\left(\frac{x}{c}\right)+\left(\frac{y}{d}\right)=1$ intersect the axes at four concyclic points and $a^2+c^2=b^2+d^2$, then these lines can intersect at $(a,b,c,d>0)$

(A) (1, 1) (B) (1, –1)

(C) (2, –2) (D) (3, 3)

Space for Rough work

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 51 and 52

Let ax + by = 1 be a chord of the curve $3x^2 - y^2 - 2x + 4y = 0$ intersecting the curve at the points A and B such that AB subtends a right angle at the origin O.

51. The value of a – 2b + 1 is equal to
(A) 0 (B) 1
(C) 2 (D) –1

52. If a and b are parameters , then $ax + by = 1$ always passes through the point (h, k), where
(A) $h + k = -1$ (B) $h + k = 1$
(C) $|h| + |k| = 1$ (D) $|h| + |k| = 3$

Paragraph for Question Nos. 53 and 54

If $\cos\alpha + \cos\beta + \cos\gamma = 0 = \sin\alpha + \sin\beta + \sin\gamma$ and $\delta = \frac{\alpha+\beta+\gamma}{3}$. The difference of any two of the angles α, β, γ is equivalent to $\frac{2\pi}{3}$. Then

53. The value of $\sin 6\alpha + \sin 6\beta + \sin 6\gamma$ interms of δ is
(A) 3 sin2δ (B) sin6δ
(C) 3 sin3δ (D) 3 sin6δ

54. If one of the angles α, β, γ is $\frac{\pi}{36}$ then the $\cos 12\alpha + \cos 12\beta + \cos 12\gamma =$
(A) 0 (B) 3
(C) $\frac{3}{2}$ (D) $\frac{2049}{2048}$

Space for Rough work